Functional Requirements

AI-Lingo: Contextual Language Immersion App

Introduction

This document lists the features and functionalities that AI-Lingo shall provide. AI-Lingo is a mobile application, built with Flutter and Dart, that uses AI to generate real-world conversational scenarios tailored to each learner's level and interests. Each requirement has a unique ID so it can be traced to user stories, use cases, and test cases.

Priorities follow the MoSCoW method: Must (essential for the first release), Should (important but not critical), Could (nice to have if time allows).

Requirements

1. User Management

ID	Requirement	Priority
FR-01	The system shall allow users to register and log in using email/password or a Google account.	**Must**
FR-02	The system shall allow users to set their native language, target language, and interests (e.g., business meetings, ordering at a café, travel).	**Must**
FR-03	The system shall allow users to edit their profile and delete their account and all associated data.	**Should**

2. Placement & Skill Assessment

ID	Requirement	Priority
FR-04	The system shall provide an initial placement test to determine the user's proficiency level (CEFR A1–C2).	**Must**
FR-05	The system shall store the user's level and update it automatically based on their performance.	**Must**

3. AI Scenario Generation

ID	Requirement	Priority
FR-06	The system shall allow users to select a scenario from a predefined list or enter a custom scenario.	**Must**
FR-07	The system shall generate a dialogue matching the user's level and interests using prompt chaining (skill-level identification followed by dialogue generation).	**Must**
FR-08	The system shall constrain generated dialogue to vocabulary the user has already learned, plus a limited number of new words.	**Must**
FR-09	The system shall highlight new words in the dialogue and display their translation and meaning.	**Should**

4. Interactive Conversation

ID	Requirement	Priority
FR-10	The system shall allow users to converse with the AI in a role-play format using text input.	**Must**
FR-11	The system shall detect user mistakes (grammar and vocabulary) and provide corrections with short explanations.	**Must**
FR-12	The system shall provide hints and on-demand translation during the conversation.	**Should**
FR-13	The system shall support text-to-speech for pronunciation and speech-to-text for voice input.	**Could**

5. Vocabulary Management

ID	Requirement	Priority
FR-14	The system shall automatically save new words learned in each lesson.	**Must**
FR-15	The system shall provide a review mode (flashcards/quiz) for saved vocabulary.	**Should**
FR-16	The system shall mark a word as "learned" based on the user's correct-answer history.	**Should**

6. Progress Tracking

ID	Requirement	Priority
FR-17	The system shall display a dashboard showing completed lessons, learned words, and the current level.	**Must**
FR-18	The system shall track daily learning streaks and send reminder notifications.	**Could**
FR-19	The system shall keep a history of past conversations and lessons.	**Should**

7. Offline Mode

ID	Requirement	Priority
FR-20	The system shall allow users to download lesson snippets for offline use.	**Must**
FR-21	The system shall store vocabulary and progress locally and synchronize them when an internet connection is restored.	**Should**

8. Settings

ID	Requirement	Priority
FR-22	The system shall allow users to change the app language, theme (light/dark), and notification settings.	**Could**

Summary

Total: 22 functional requirements across 8 modules. Must: 12, Should: 7, Could: 3.